

Huawei Full-Stack Data Center, Powering Industry Intelligence

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Trends and Challenges



Industry Overview: Technology Evolution in Data Center

Transformations in 2025



Smart Meters

30M

2023 15M



Charging Stations

1M

2023 <20万



Terminal
Units

>200M



Data Growth

30%

CAGR

**Data: EO Intelligence Forecast*

Industry Trends in Data Center



Industrial
Computer **OT**

OT Core Production

Maintaining Distributed Deployment

Traditional architecture continues to exist:

- Core production systems maintain distributed silo-style deployment
- Business security and fault isolation



Management
Computer **IT**

IT Non-Production and New Business

Attempting Cloud Integration

New business scenarios rapidly evolving:

- Frequent user interactions, widespread video surveillance, and explosive data growth.
- AI applications starting pilot programs from internal IT systems

Cloud: Foundation for Industry Digitalization

Business Challenges

Silo Construction Dilemma



- Scattered and fragmented systems, difficult to support unified group control
- Low resource utilization, high TCO
- Fragmented IT infrastructure, high operational complexity

Digital Transformation Pressure



- Urgent IT/OT integration trends, need to find breakthrough points
- Gap between AI and other new technologies with traditional architecture

Dual-track Data Center **Architecture** Balances Stability and Innovation

Cloud-based Data Center

OT

Silo & Status Quo

AMI



CCTV



IT

Silo & Integration

IT Systems



System integration is an inevitable requirement for advanced data center architecture.

The urgent priority is to **start integration with non-production IT systems.**

1 Unified Cloud Architecture

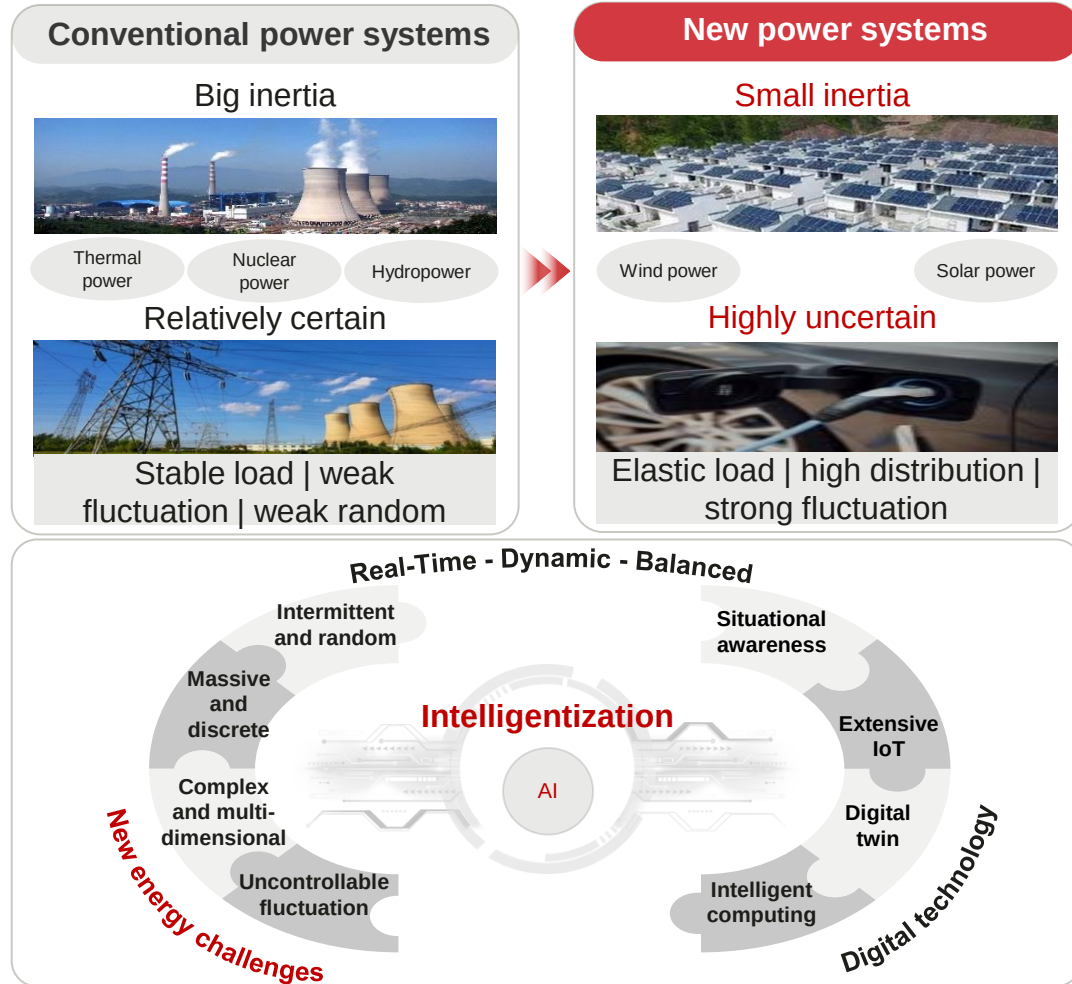
2 Security and Control

3 Smooth Evolution

4 Enhanced O&M

AI: Driving Force for Industry Digitalization

High Uncertainties



Major Applications and Researches

Power asset management and intelligent O&M

- Evaluation and diagnosis of device health status
- Troubleshooting assistance

New energy consumption

- New energy power generation forecasting
- Collaborative control of power supply, power grid, load, and energy storage

Perception and prediction of new types of loads

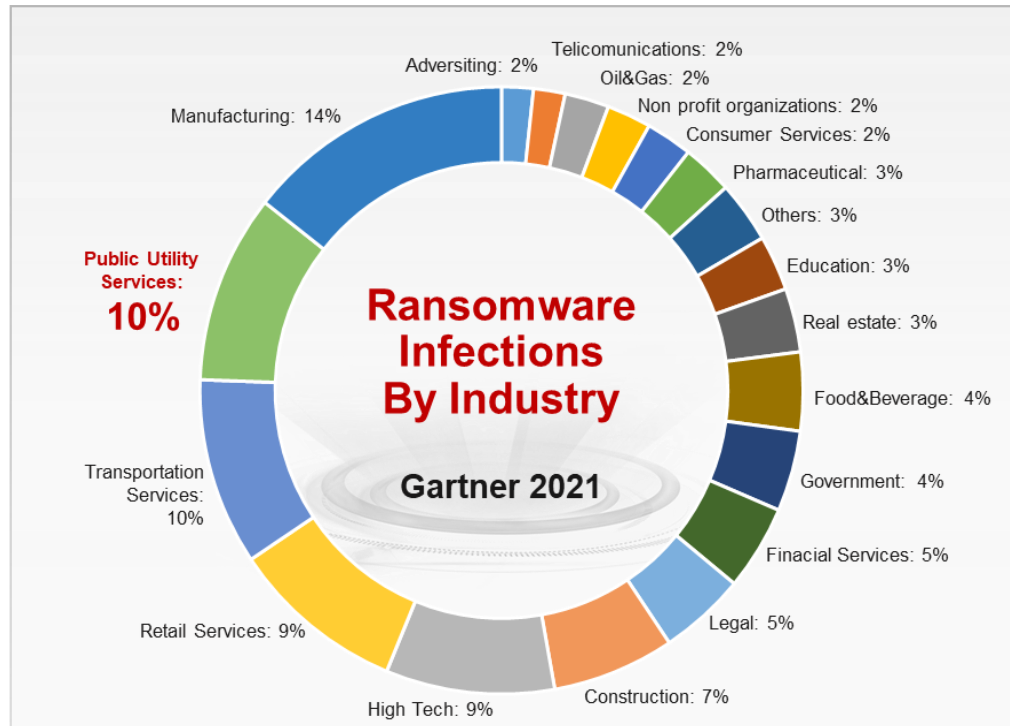
- Charging behavior analysis and demand response development
- Effective allocation and energy consumption guidance

Secure and stable power grid

- Quick stability determination, fault identification and prevention, and emergency control measures

Data Security: Cornerstone for Industry Digitalization

Data Security Landscape



The power industry ranks second among sectors most targeted by ransomware attacks

Secured Data Center

Cyber Attack



Virus & Ransomware



Growing Access Points and Network Exposure

Critical Infrastructure Vulnerabilities:

- Production: SCADA/EMS stability, blackout risks
- Marketing: AMI systems, data breach threats
- IT: Enterprise operations disruption

Ukraine/Venezuela blackouts, \$14M+ ransomware demands

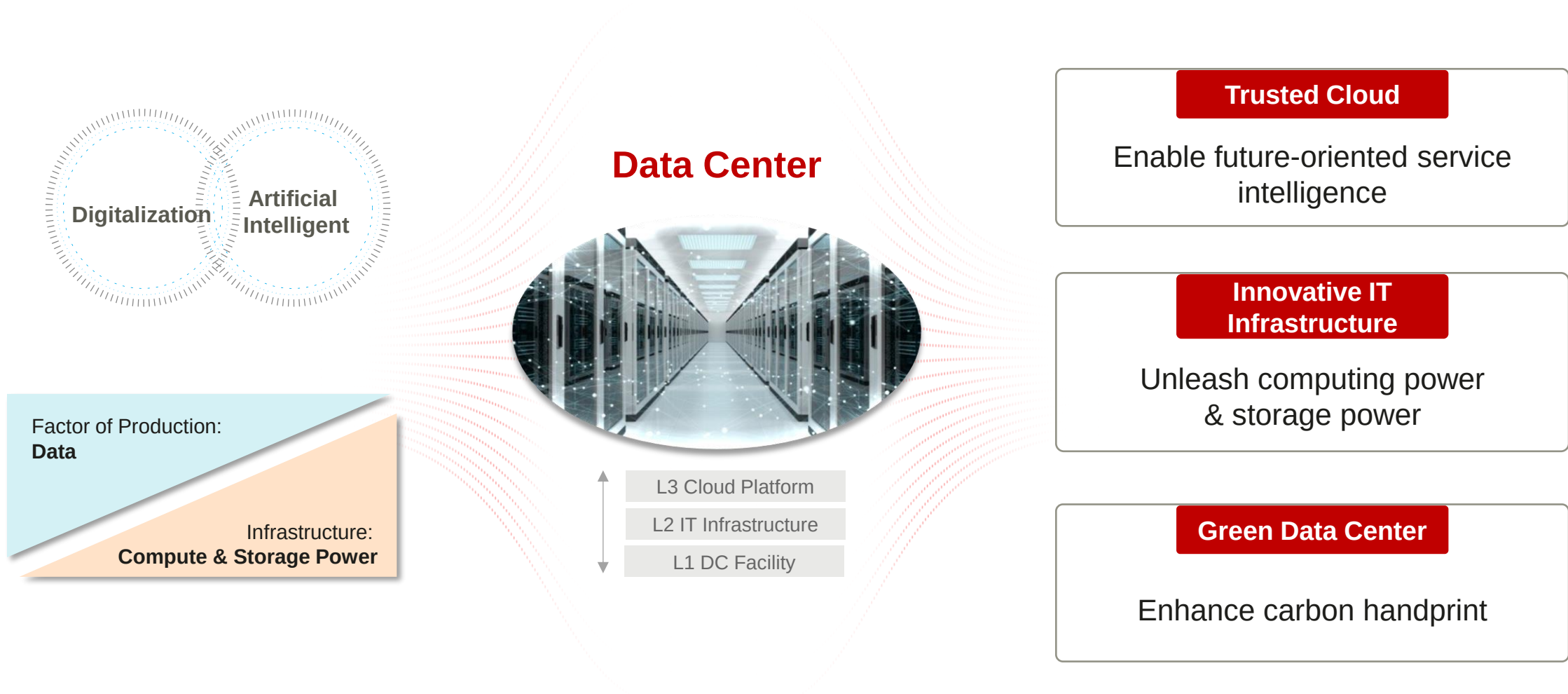
Security Requirements

- OT/IT security hardening: critical operational and business data integrity
- Data full lifecycle protection: data security from creation to disposal
- Cloud access control: securing services and resources
- Network boundary protection: safeguarding expanded network perimeter

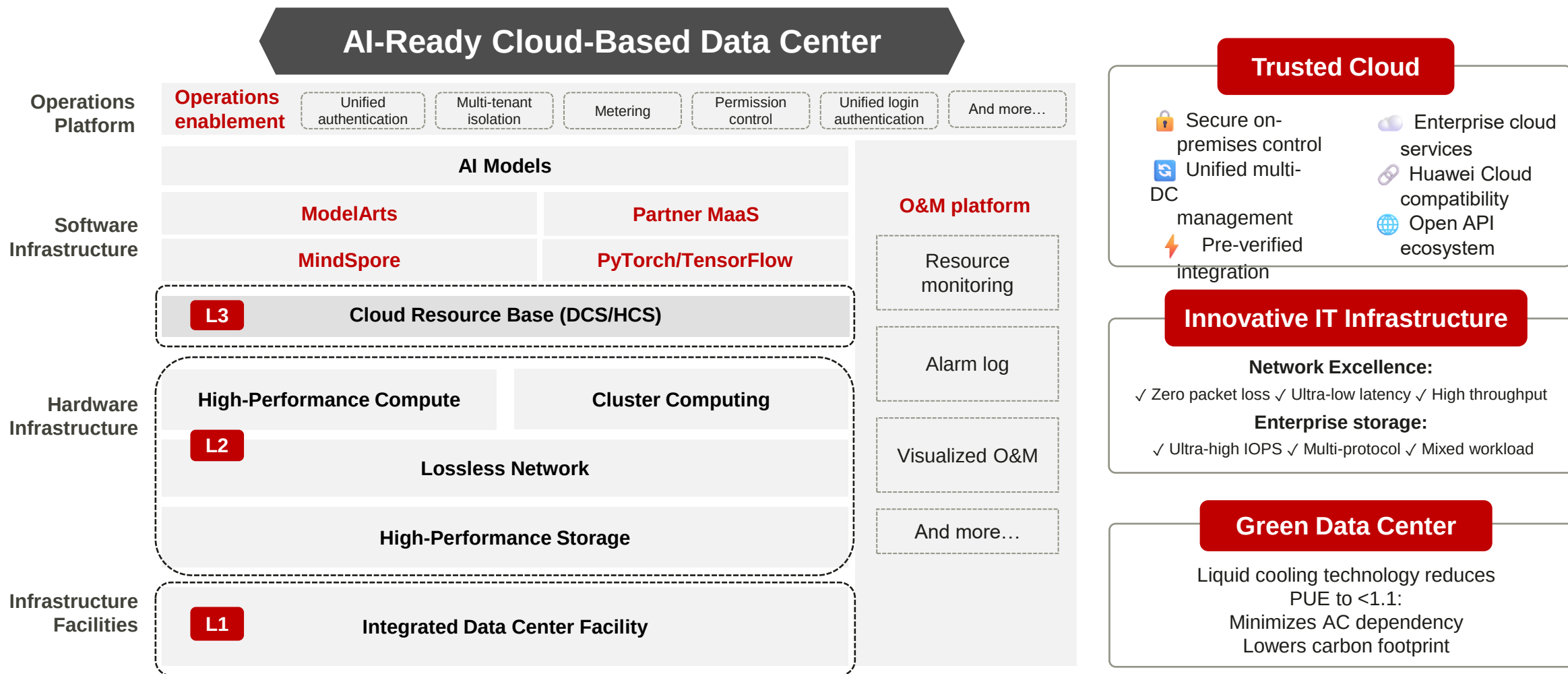
Huawei Solution

Huawei Full-Stack Data Center

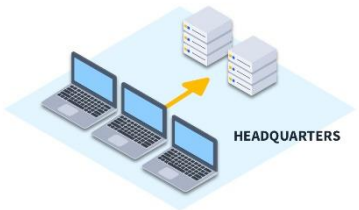
Data Centers: The Digitalization & AI Foundation



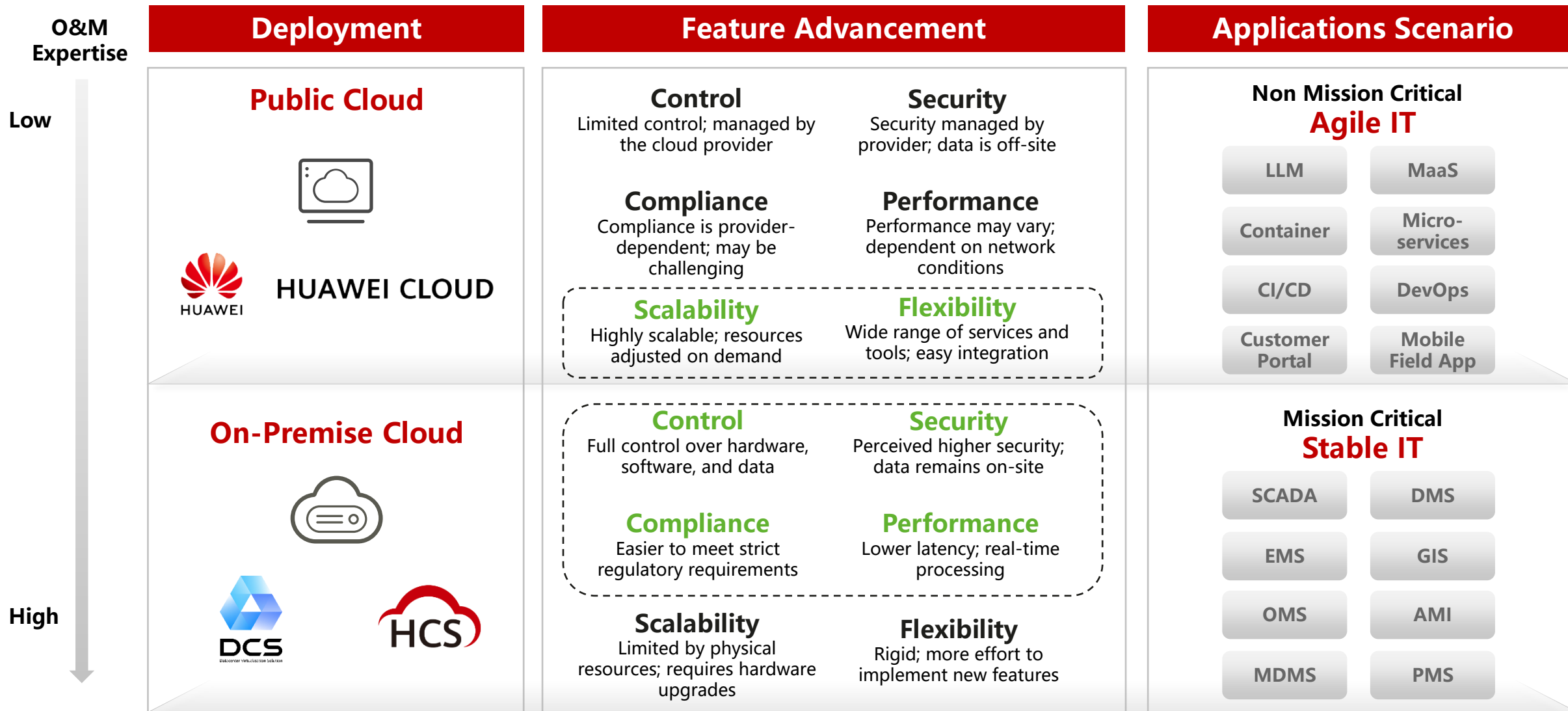
Huawei Full-Stack Data Center Technical Architecture



Facility Selection: On-Premises vs Off-Premises Data Center

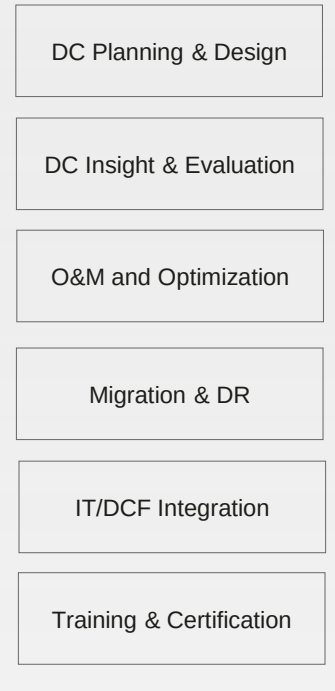
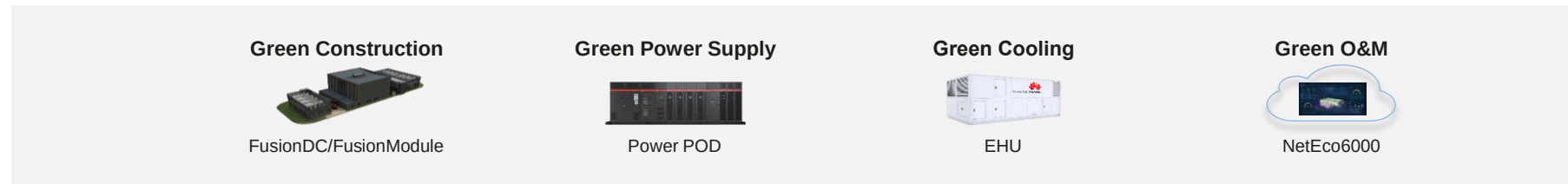
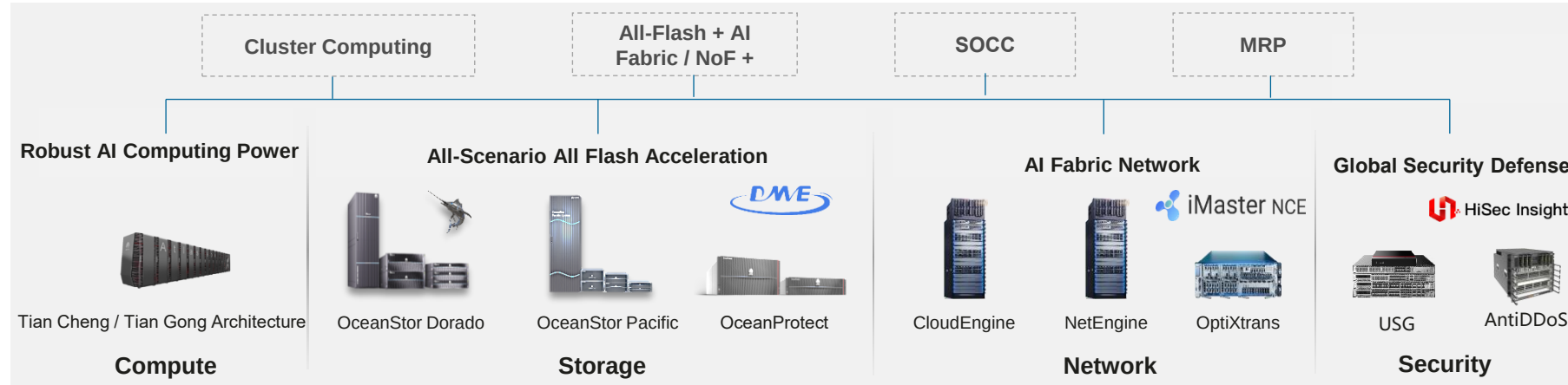
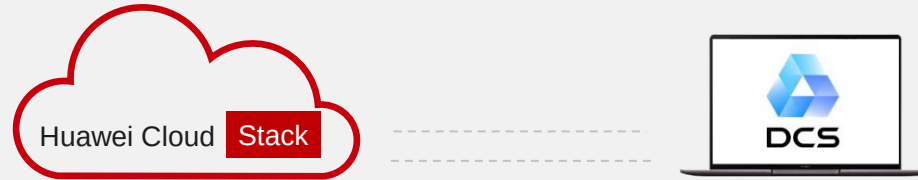
Data Center Facility	Advantages & Disadvantages	Recommendations
On-Premises (Self-Built)  <i>High-standard, customized facilities</i> <i>Full ownership and operational control</i> <i>long-term strategic value</i>	Pros CONTROL: Full lifecycle management from design to operations, enabling customized specifications and autonomous decision-making RELIABILITY: Enhanced security, unrestricted access, and direct operational visibility STRATEGY: Platform for building business ecosystem and strengthening corporate brand value Cons INVESTMENT: Significant upfront capital expenditure and long-term financial commitment TIME: Extended project timeline from planning to operational readiness	Mission-critical core businesses SCADA DMS EMS GIS OMS AMI MDMS PMS
Off-Premises (Colocation)  <i>Standard colocation facilities</i> <i>Optimize for operational flexibility</i>	Pros FLEXIBILITY: Fast deployment with minimal upfront investment and scalable capacity SIMPLICITY: Reduced operational complexity and management overhead Cons LIMITATION: Restricted control over design, operations and infrastructure decisions DEPENDENCY: Reliance on provider for security, access and service quality	Non-critical supporting services CIS CRM OA ESB OCS MIS

Platform Selection: Public Cloud vs On-Premise Solutions



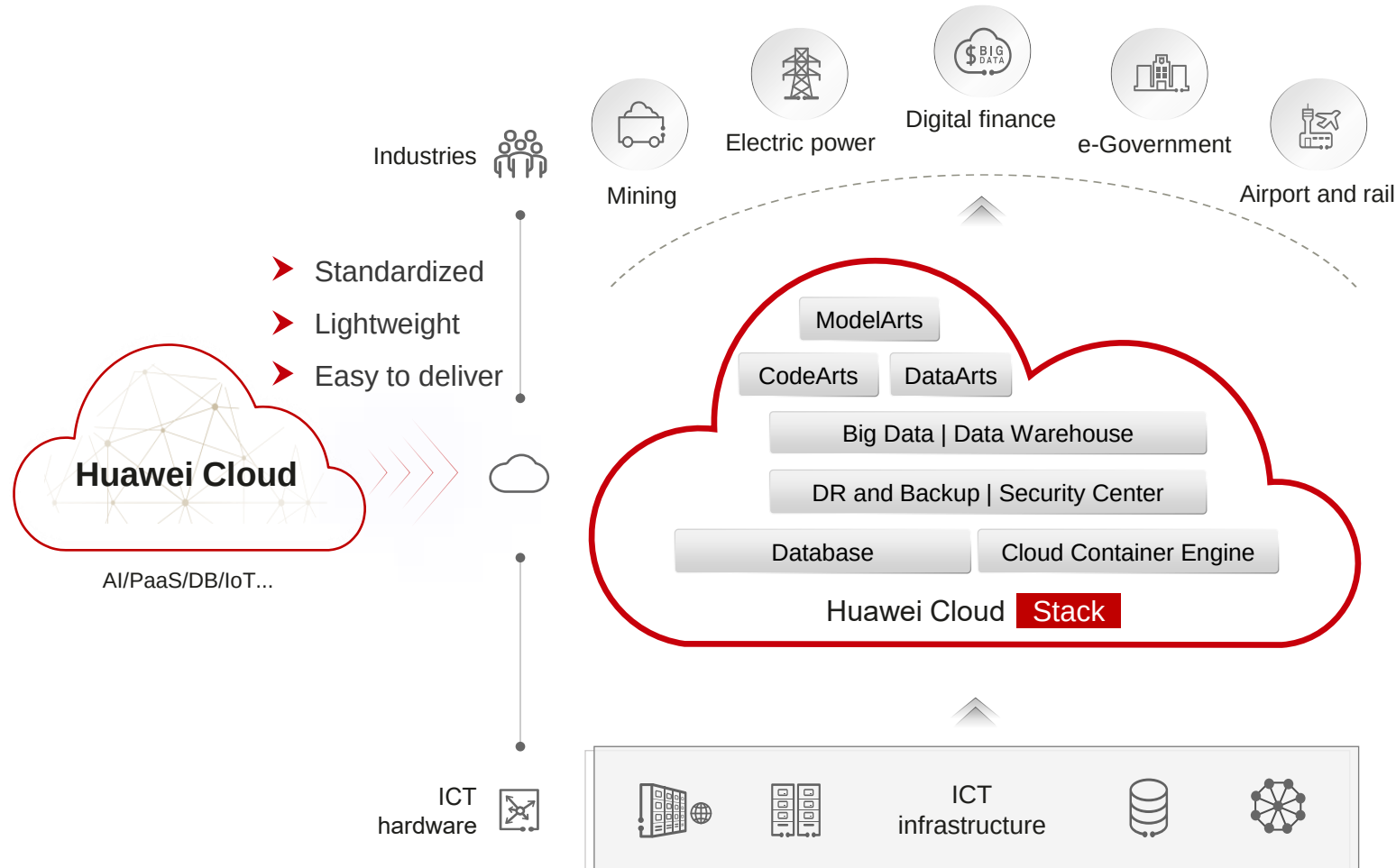
Leading Data Center Infrastructure, Powering Industry Intelligence

Huawei Data Center Products & Solutions



Trusted Cloud

Huawei Cloud Stack: A Trusted On-Premise Cloud Solution



On-premises deployment

- **Secure:** controllable data and legally compliant data use
- **Local O&M:** unified O&M of multiple data centers
- **Collaborative:** hardware and software pre-integrated and verified before delivery

Comprehensive cloud services

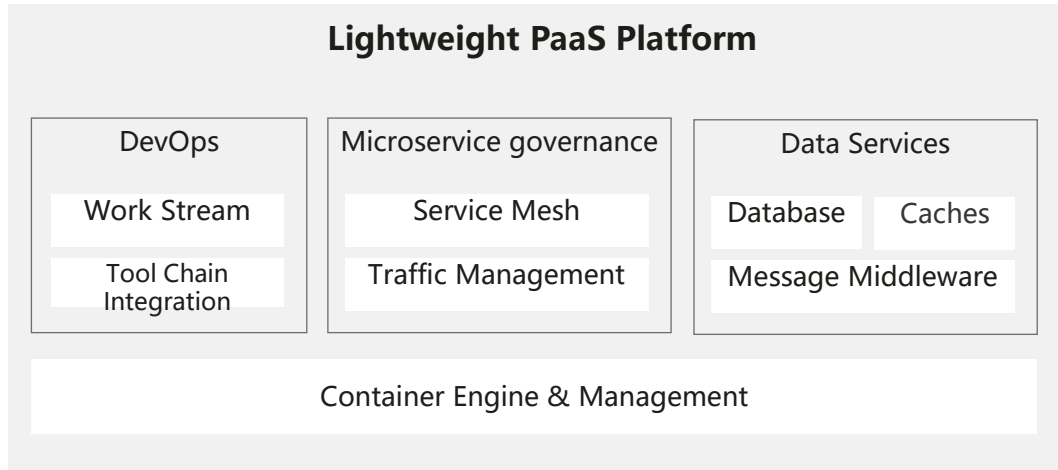
- **Unified architecture:** consistent experience with Huawei Cloud
- **Keep evolving:** continuously inherit cloud services from Huawei Cloud
- **Open:** extensive ecosystem and standard APIs

Multi-cloud management

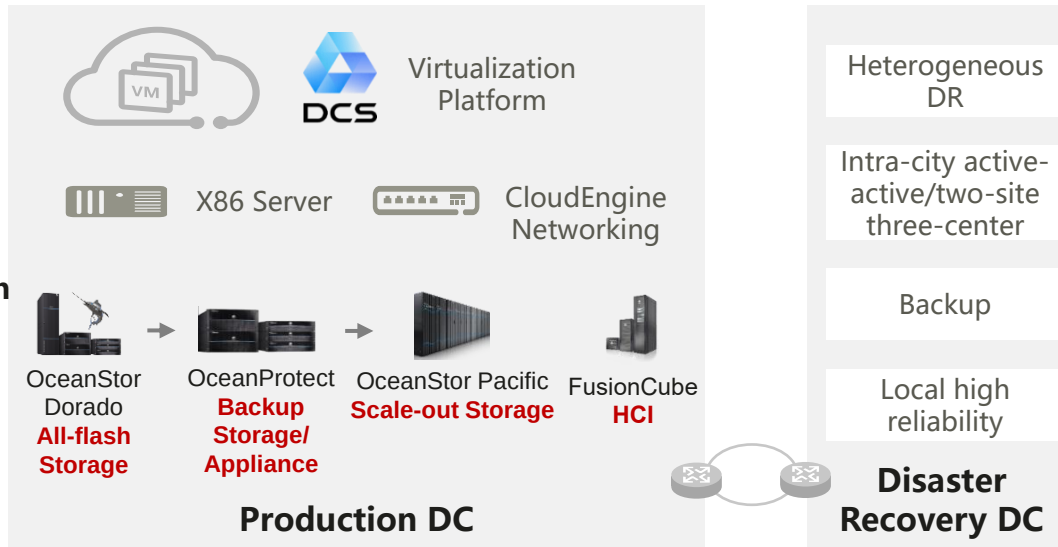
- **Unified access:** 10+ resources from multiple clouds
- **Automated orchestration:** Service Builder enable a faster rollout (weeks → hours)

LightDC: Lightweight Digital Base on Huawei DCS

PaaS Platform



Virtualization Base



Data center management

Resource orchestration

Intelligent O&M

Multi-DC management

Multi-tenant operation

Large-screen display

Mgmt.



Security situational awareness



USG Firewall



Sandbox



EDR

Security



Ultimate Agility - Minimum of 2 nodes
Lego-like Architecture



Ultimate Resilience -
6-layer ransomware protection by network-storage-compute collaboration



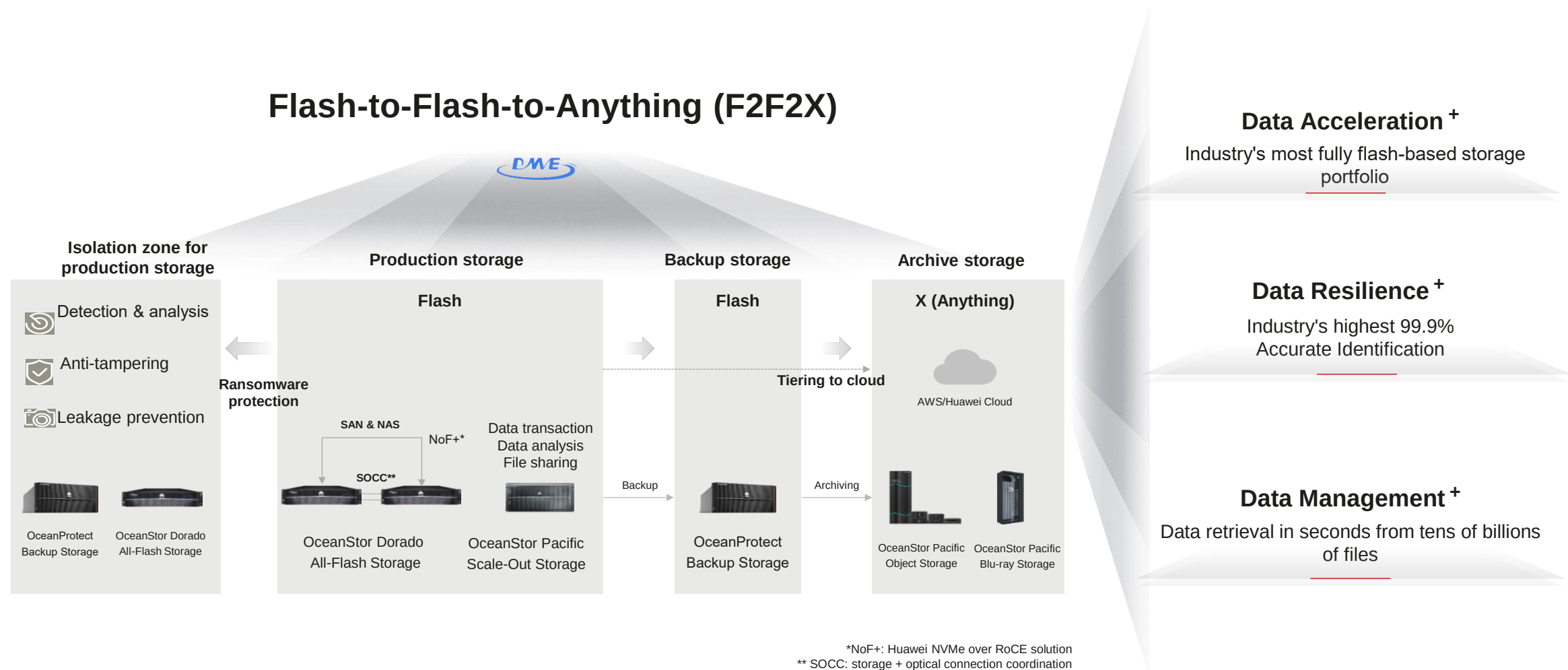
Ultimate Reliability
- All-domain and all-scenario DR and backup



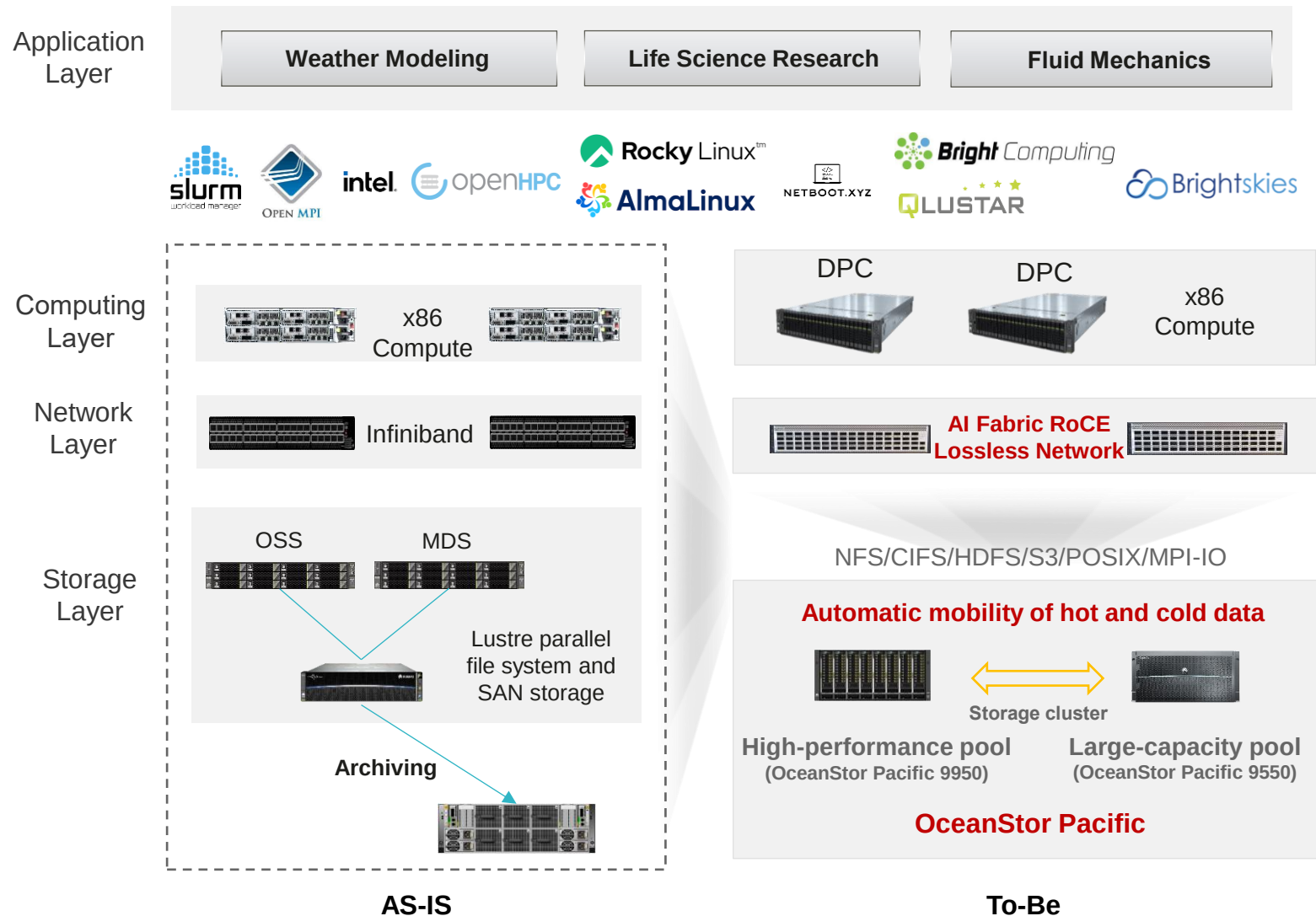
Simplified O&M
- Unified datacenter management

Innovative IT Infrastructure

F2F2X Data Infrastructure Unleashes All-Flash Storage Power



Smart Network: Powering Supercomputing



Fast Delivery Accelerates Service Rollout

Huawei OceanStor Pacific (DPC)+RoCE solution to replace Lustre+IB solution, improving delivery efficiency.



Future-Oriented Unified Platform Reduces TCO

Pacific provides seamless multi-protocol interworking, enabling access to multiple services using different protocols.

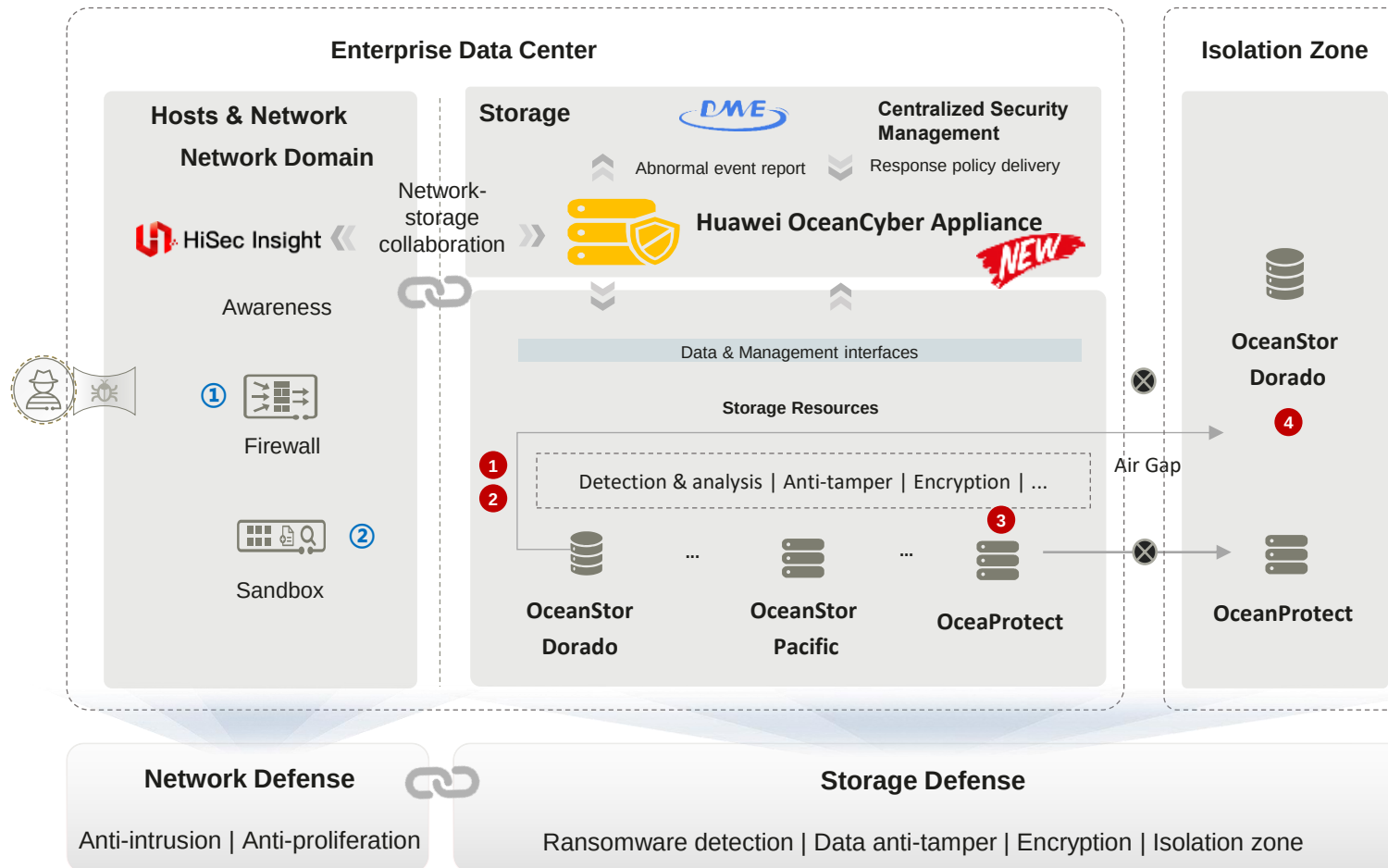
AI Fabric provides a converged all-Ethernet architecture.



Assured Performance for Diversified Service Workloads

Enterprise level DPC+RoCE solution competitive with Lustre+IB solution, meeting the requirements of diversified service workloads.

Data Security: Next-Gen Data Security Shield



Industry-Leading 99.9% Detection Accuracy

Ransomware detection
pre-event, during-event & post-event

Multilayer Comprehensive Protection

- ① Network Anti-intrusion
- ② Network Anti-proliferation
- ① Detection
- ② Secure Snapshot
- ③ Backup
- ④ Air-Gap

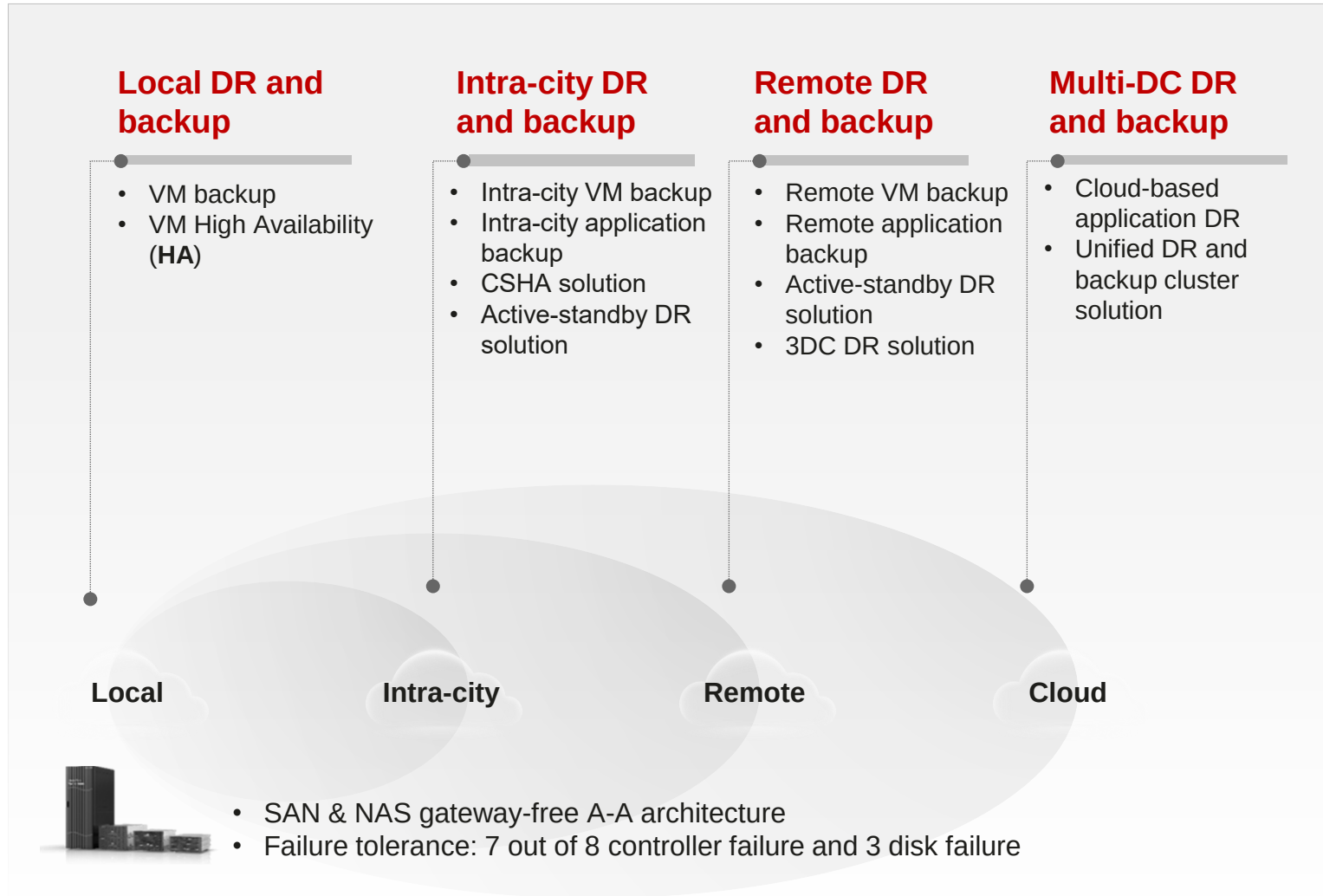
Industry's Fastest Recovery

Secure snapshots: seconds-level recovery
OceanProtect: up to 172 TB/hr recovery

Industry's 1st Network Security-Storage Integration

The network quickly notifies the storage once a ransomware is detected; Storage takes immediate action to protect data

Ultimate Reliability: 4-Level DR and Backup Solutions for All Scenarios



Extensive App Protection

Wide range protection of applications



Multi-Region DR and Backup

RPO = 0, RTO in seconds

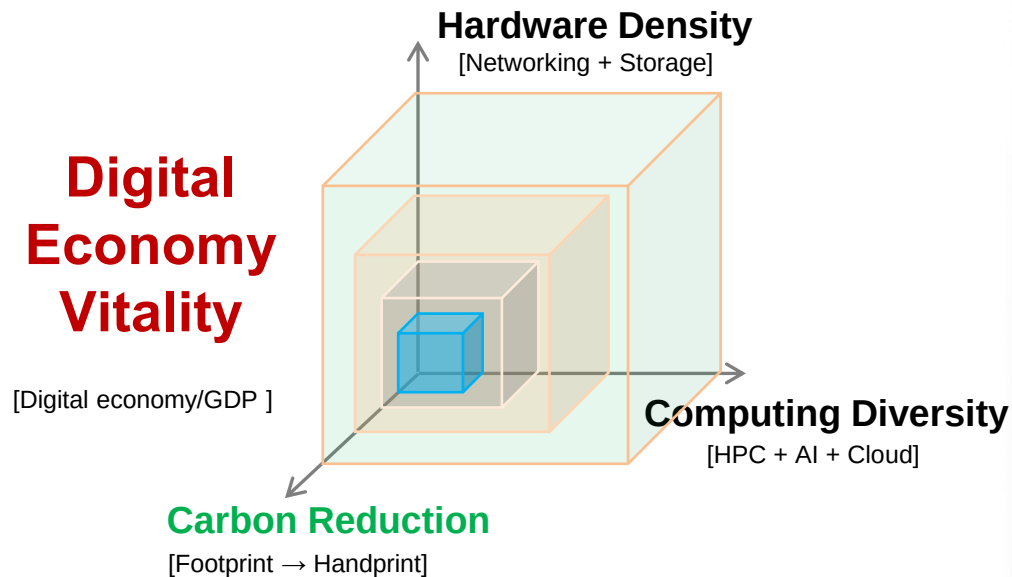


Cost-Effective DR and Backup

65:1 Deduplication Ratio

Green Data Center

Carbon Reduction is Crucial for Sustainable Development



“The strength of the digital economy is roughly equal to connectivity density times computing power. Now we need long-term vitality. So, we need to **consider a new dimension, carbon reduction**”

Huawei Chairman, MWC 2022

“More Throughput, Less Watts”

Carbon Footprint



ICT's own footprint of 1.25Gt CO₂e in 2030

Enablement
~10x



Source: GeSI / GSMA

Carbon Handprint



Emissions avoided through use of ICT 12Gt CO₂e

“Successful implementation of Industry 4.0 transformation can simultaneously augment green measures while bolstering production efficiency. Thus, achieving **eco-efficiency**, wherein **sustainability** and **competitiveness** are not only compatible but, in fact, are **interwoven**”

Source:  

Data Center L1 Facility Challenges

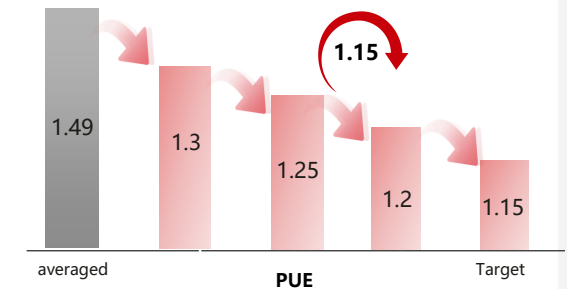
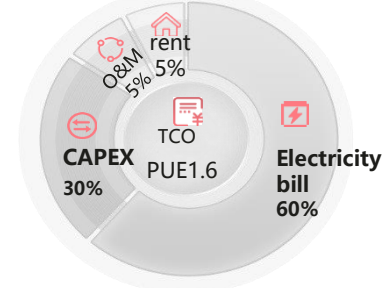
Difficult to adapt to new computes such as AI, HPC

The infrastructure lacks density and flexible, fail to cope with the uncertainty of new computing powers such as AI and HPC

XPU	150-350W (x86/Arm)	500+W (x86/Arm)	5kW (Nvidia/Ascend)
Rack	5-8 kW/cabinet	20-66 kW/cabinet	
Power & Cooling	General Infrastructure	New Infrastructure	

High energy cost, PUE fails to meet the target

10-year electricity bill =
2 new data centers (L1)

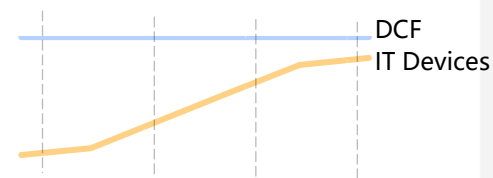


Long construction time, unable to match IT evolution

Long construction period,
TTM > 18 months



One-off investment,
unable to deploy on-demand



High O&M costs and low O&M efficiency



O&M personnel
shortage

47%

Low resource
utilization

< 65%

Low SPC usage

Modular Data Center: Smart Modular DC 6.0 One-stop Solution

Traditional Solutions

Traditional data center solutions cannot support high density and future service evolution.

Challenges



Density

Traditional DC architecture supports **3–5 kW/rack**



Design

Room-level air conditioners are used for air supply



Reliability

Hot spots are easily generated

FusionModule 6.0



Modular

One-stop delivery
On-demand deployment

Green

30% (1.6 → 1.111)
The PUE is reduced by 30%

Simple

30 days → **1 week**
Minimal deployment

Reliable

Cooling "0" interruption
New architecture for continuous cooling

Complete EPC

Flexible expansion
Integrated UPS/Space
saving

Three key technologies

Extreme energy saving
Fluorine pump/Full adaptive
compressor/AI

Low footprint

(75.6m²→64.8m²)

Low- Room Height

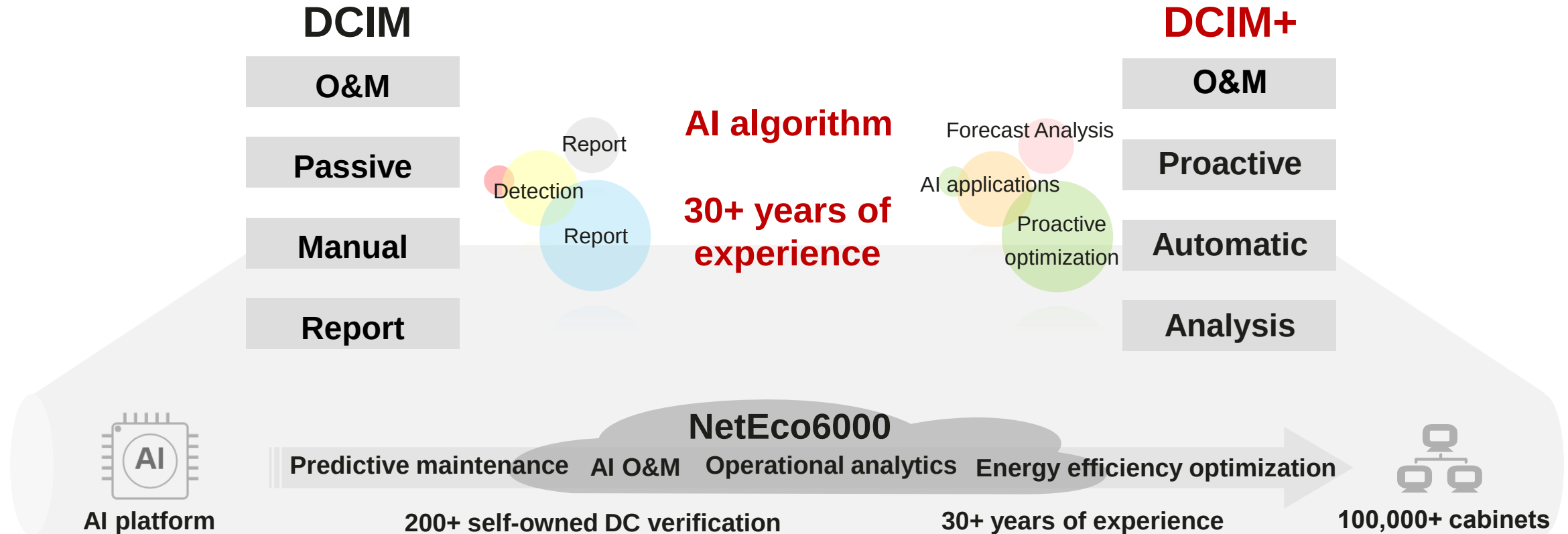
(3.0m→2.6m)

Free of raised floor

Continuous cooling

Ultimate Reliable
Device security/Architecture
security

DCIM Evolution: AI-Driven | 30+ Years O&M Experience



AI-Powered NetEco6000 Solution Overview

AI energy optimization

Reduces data center energy consumption by 8 to 15%.

Automatic maintenance

Reduce maintenance labor cost by 35%

Intelligent operation

Increase data center resource usage 20%

Security

Ensure network security

Energy

1 Visualization

Energy consumption visualization

2 Diagnosis

Diagnosis and optimization

3 Adaptation

Continuous optimization based on scenarios

4 Optimization

AI-based intelligent optimization

Maintenance

- Digital App O&M
- iPower preventive Maintenance

Operation

- Capacity Optimization
- Asset Auto-Discovery



Security

(End-to-end security)

AI Platform

iCooling collection platform
iCooling inference platform



Data Collection



Model Training



Model Inference

Data center infrastructure



Power distribution device



Cooling device



Smoke



Temperature & Humidity



Air pressure



Data collector



Video & Door access

Full-Lifecycle Service

Full-Lifecycle Professional Data Center Services



Blueprint Design



Planning, design, and
implementation



Integration and
development



O&M



Training and
certification

One-stop, global service

Asia Pacific, Latin America, South Africa, Middle East...

Based on Huawei's **30-year** 2B service experience and global service layout

One-stop global service

Expert system

170 +

Global Service Centres

30 years

2B service experience and global service layout

20 +

Service tools, service specifications, and
service methodologies

2000 +

O&M Experts

4000 +

Service Engineers

1000 +

Global Services Partners

10000 +

Certified Service Engineers

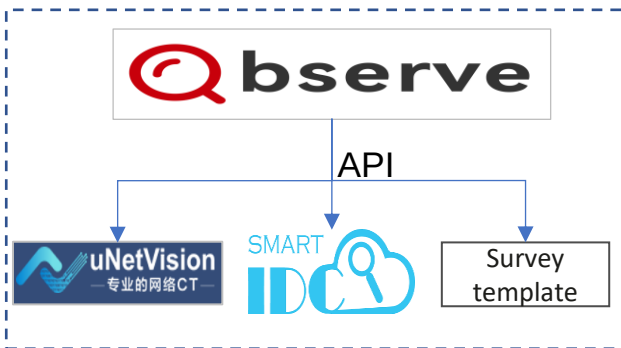
Smart Data Center Assessment: Automated diagnostics, Expert solutions, Strategic planning

Simplify Investment Decision-making, Improve DC Infrastructure Health, and Optimize Planning Quality

Analyze the current DC status

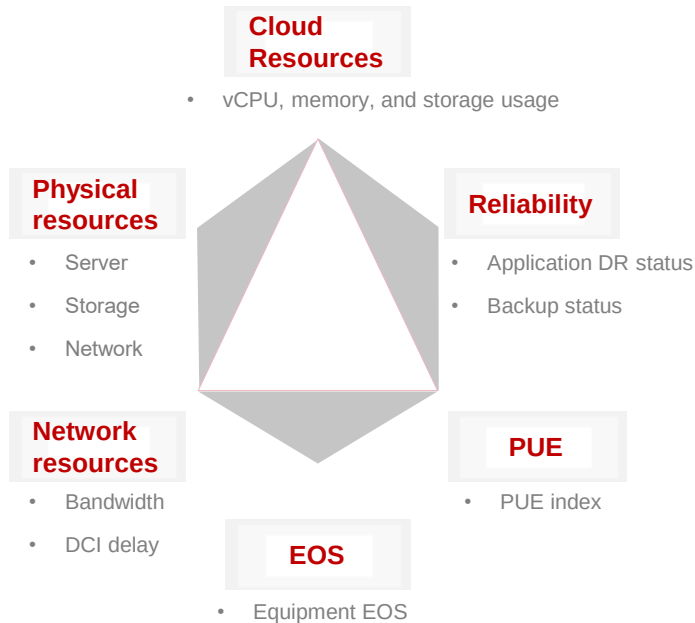
Integrate multiple tools to insight ICT infrastructure of Data Center

Use tools to automatically collect device logs and manually supplement survey tables to obtain detailed insights into the live network infrastructure of the data center.



Find pain points

Identify potential problems from six dimensions



Output suggestion & report

Generate analysis report and provide optimization suggestions

Systematically summarize the DC status and provide a network evaluation report.



Thank you.

Bring digital to every person, home and organization for a fully connected, intelligent world.

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